Detecting Personal Data in Chatbot Conversations: Pretrained NER Models on a Synthetic Benchmark, with an Address-Coverage Evaluation

1. INTRODUCTION

National statistical institutes (NSIs) run chatbots for contact centres and survey support, and chatbots are among the most common AI applications in the public sector [1]. Users type names, addresses, phone numbers and codes in these conversations. Reusing the text for analysis requires anonymisation, within a statistical disclosure control framework and in line with the GDPR [2].

Early anonymisation used rule-based methods, which are precise on identifiers with a fixed format and weak on context-dependent information; later studies confirm their limits on heterogeneous text [3]. Medlock introduced a probabilistic approach and separated direct identifiers, such as names, from quasi-identifiers, such as organisations or contextual references [4]. Much of the work comes from clinical de-identification [5, 6]. Recent systems use Transformer models [7] and often combine rules with neural models [8]. A second line comes from privacy-preserving data publishing. Removing explicit identifiers does not remove the risk, because quasi-identifiers and rare combinations of attributes remain [9], and anonymisation can be defined as a transformation with explicit privacy guarantees [10]. The two lines are rarely joined for free text [11].

A first pipeline for the chatbots of the Italian National Institute of Statistics (Istat) combines regular expressions for structured identifiers with a fine-tuned multilingual BERT for names and locations [12]. It was trained on synthetic sentences and evaluated on 300 manually annotated real conversations (sensitivity 0.966, precision 0.704). That evaluation has two limits. The gold standard is built on real chats, so it stays inside the institute, it is costly to extend and it cannot be used to compare many off-the-shelf models. And span-level metrics say how many entities were found, not how much of a person's identity stays visible. This matters most for addresses, which have a street, a number, an internal and a municipality: a model that finds only the municipality scores a hit and leaves the street and the number exposed.

This paper addresses both limits, with three contributions. First, we build a synthetic benchmark of 289 institutional chatbot conversations with a programmatic gold standard, which needs no real conversations and can be regenerated and extended. Second, we compare four pretrained NER models, without fine-tuning, on PERSON and LOCATION. Third, we add an address-coverage measure that does not depend on how a model splits an address into spans, and we report bootstrap confidence intervals.

2. METHODOLOGY

2.1. Synthetic dataset

The catalogue has 12 scenarios derived from anonymised chatbot conversations, for example access recovery, form-filling support and change of residence. We generated 300 conversations: 230 from the general scenarios, 10 to rebalance five

central scenarios and 60 as an address stress test. In total, 175 contain personal data and 125 do not.

Generation splits three tasks. The code chooses scenario, style and values. The language model (gpt-5.6-terra) writes only the dialogue, using authorised placeholders. The code then replaces the placeholders with synthetic values, computes the offsets and builds the gold standard. Names, surnames, streets and municipalities come from local catalogues; house numbers, internals and other values follow versioned rules. The model produces no annotations, so it does not decide what counts as correct.

Automatic checks on schema, slots, offsets, turn alternation and coherence validated 289 conversations (96.3%). The 11 excluded ones are kept with the reason and were not corrected. The benchmark has 123 PERSON and 296 LOCATION annotations. The address stress test uses six surface forms, ten prompts each: full (*Via Angelo Lauri 96, Anzio*), without street type, without house number, without municipality, with complex number (*96/A*) and with internal. The benchmark contains 124 address occurrences. A repeated address counts again, because an unrecognised repetition leaves the value exposed.

The gold allows overlapping annotations. An address is stored as a whole ADDRESS and, in the NER view, as one LOCATION for street and number and one for the municipality. The components are produced during generation and not inferred afterwards.

2.2. Workflow and models

Figure 1 shows the workflow. Regex/lookup and NER read the original text independently, and each detection keeps its provenance. A later step merges compatible detections and keeps overlaps and conflicts visible. A conversation-level policy then decides on MASK, KEEP, GENERALIZE or REVIEW. The experiments evaluate only the NER branch, at message level.

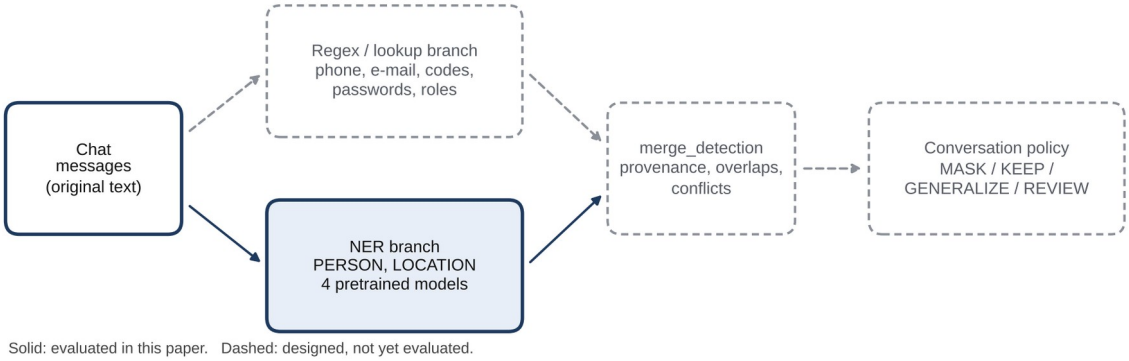


Figure 1. Detection and masking workflow. Only the NER branch is evaluated here.

We compare GLiNER multi, GLiNER PII [13], WikiNEuRal [14] and Piiranha [15] on the same messages, one at a time, with no fine-tuning and no access to the gold or to the conversation. GLiNER multi receives the labels *person* and *location*. GLiNER PII also receives *address*, mapped to LOCATION. Both use threshold 0.5. WikiNEuRal uses PER and LOC. Piiranha uses GIVENNAME and SURNAME, merged into PERSON, and STREET, BUILDINGNUM and CITY: STREET and

BUILDINGNUM are recomposed into one LOCATION and CITY stays separate. Thresholds were not tuned on the 289 conversations. The label mapping is part of the protocol: a first run that left out STREET and BUILDINGNUM gave Piiranha 0/124 complete addresses, and we do not use it. The weights of WikiNEuRal and Piiranha have non-commercial licences, to be checked before any production use.

2.3. Evaluation

Exact match counts a prediction as correct when type, message and span boundaries equal the gold, as in the span-level scoring of the CoNLL shared tasks [16]. We report micro precision, recall and F1. Address coverage is the share of the alphanumeric characters of an address covered by the union of the predicted LOCATION spans. An address is complete when all its characters are covered. With the union, one span for the whole address and two spans for street and municipality are equivalent. We also report coverage by component. Confidence intervals are 95% percentile intervals from 2,000 bootstrap resamples of conversations.

3. Results and Practical Application

Table 1 gives the exact-match results. Piiranha has the highest F1 overall (0.876) and on both categories. Its overall interval does not overlap with that of any other model; on LOCATION it overlaps with GLiNER multi. It produces no false positive on the 163 conversations without these two categories. GLiNER multi has the highest recall (0.924) and 445 false positives. Of these, 350 are PERSON spans, mostly role words (*referente* 143 times, *precedente referente* 36 times). The model confuses a person with a role. The count is inflated by repeated templates in the synthetic data and is not a prevalence in real traffic.

Table 1. Exact-match results on 289 conversations; 95% bootstrap intervals in brackets.

Model	Precision	Recall	F1	PERSON F1	LOCATION F1
GLiNER multi	0.465	0.924	0.619 (0.559–0.675)	0.397 (0.328–0.465)	0.817 (0.763–0.865)
GLiNER PII	0.198	0.465	0.278 (0.235–0.318)	0.342 (0.280–0.404)	0.213 (0.149–0.281)
WikiNEuRal	0.633	0.666	0.649 (0.600–0.698)	0.763 (0.686–0.832)	0.599 (0.539–0.661)
Piiranha	0.865	0.888	**0.876 (0.833–0.919)**	**0.911 (0.857–0.960)**	**0.862 (0.807–0.916)**

Table 2 shows address coverage, and the ranking changes. GLiNER PII covers 122 of 124 addresses completely and GLiNER multi 120; the difference between them is within the intervals. Piiranha covers 92 and misses all 17 internals. WikiNEuRal covers 16: it finds street and municipality but leaves the house number visible (4 of 110 complete).

Table 2. Address coverage on 124 occurrences. Components are counted when fully covered.

Model	Complete	Rate (95% CI)	Coverage	Street	Number	Internal	Munic.
GLiNER multi	120	0.968 (0.928–1.000)	0.983	121/124	107/110	17/17	105/105
GLiNER PII	**122**	**0.984 (0.960–1.000)**	**0.986**	122/124	108/110	17/17	103/105
WikiNEuRal	16	0.129 (0.056–0.222)	0.792	111/124	4/110	0/17	98/105
Piiranha	92	0.742 (0.631–0.852)	0.915	109/124	102/110	0/17	105/105

Exact match and coverage disagree most for GLiNER PII. Its LOCATION F1 is 0.213 because it often returns the full address as one span, for example *Via Casal Di Zaino 109, Cisterna di Latina*, while the gold has two. For masking the two forms are equivalent. The cost is elsewhere. GLiNER PII marks 9,016 characters outside the PERSON–LOCATION gold, including e-mail addresses that the regex branch

should handle, and it produces false positives on 123 of the 163 conversations without these categories, for example on *assistenza* and *recapito*. A wide span helps when it covers an address and harms when it masks a generic word. Table 3 shows the text left visible for one address of each form.

Table 3. Address text left visible by each model; one example per form, chosen where GLiNER multi covers the address and WikiNEuRal does not.

Address	GLiNER multi	GLiNER PII	WikiNEuRal	Piiranha
Via Casal Di Zaino 109, Cisterna di Latina	none	none	109	none
Giovanni Agnelli 185, Minturno	none	none	Giovanni Agnelli 185	Giovanni Agnelli
Traversa Prima Vigne Inferiori, Santi Cosma e Damiano	none	none	Traversa Prima Vigne Inferiori	none
Via Colle Sant'Upica 28	none	none	28	none
Via Silvio Pellico 198 bis, Antrodoco	none	none	198 bis	none
Via La Foresta 169, interno 7, Anagni	none	none	169, interno 7	interno

4. MAIN FINDINGS

No single model is best on all counts. On this corpus Piiranha is the most balanced detector for PERSON and LOCATION, and the two GLiNER models protect addresses best at a high cost in false positives. Three findings follow. First, exact match alone misleads for addresses: it puts GLiNER PII last on LOCATION F1, while that model covers 98% of the addresses. Second, the label configuration is part of the result: the Piiranha mapping moved complete addresses from 0 to 92. Third, the component view shows which part of an address stays visible, the house number for WikiNEuRal and the internal for Piiranha, and the aggregate metrics hide it.

In practice the results support a composite pipeline: Piiranha for PERSON, a high-recall address detector from the GLiNER family followed by a filter for role and generic words, and regex for structured identifiers, with a merge step that keeps provenance. We have not yet measured whether this combination improves the final result.

The results hold for a synthetic corpus with a programmatic gold that no human annotator has reviewed. Addresses are oversampled by design, each model has a single configuration and no real conversation is involved. The next steps are to implement and evaluate the merge and the role filter, to calibrate thresholds on separate development and test sets, to build a small human-annotated gold, and to add the conversation-level policy. We then plan to move from detected entities to re-identification risk, including combinations of quasi-identifiers within a conversation [9, 10], which connects to the disclosure risk indicator of [12]. Tests on real conversations will take place in the institute's own environment.

5. REFERENCES

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